

Quantitative Options Arbitrage: Matrix Transformations and Constrained Optimization Framework

Systematic Trading Research

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Abstract

This report outlines the mathematical formulation for a statistical relative-value options trading strategy. The framework is divided into two primary components: (1) the transformation of raw Level-5 tick data into an actionable mispricing score vector $\mathbf{W}$, incorporating strict liquidity filters, and (2) a constrained convex optimization model designed to maximize the portfolio's expected alpha (dot product of positions and scores) while enforcing strict exposure bounds.

1 Part 1: Pricing Matrix Transformation and Scoring

The objective of this phase is to evaluate the current market state across N available option contracts and generate a score vector $\mathbf{W} \in \mathbb{R}^N$ representing the degree to which each option is overpriced or underpriced.

1.1 Liquidity Filtering and Effective Price Matrix

Raw L5 order book data contains multiple levels of bids and asks, but not all are tradable due to liquidity constraints. Let $i \in \{1, 2, \dots, N\}$ index the available options. At any tick t , we define the *effective* ask price P_i^A and effective bid price P_i^B using volume thresholds.

Given the actionable latency bounds, we accept the first level ($L1$) if the volume $v(\cdot)$ exceeds 5 contracts. If not, we sweep the first two levels ($L1 + L2$) if their combined volume exceeds 5. Otherwise, the option is deemed untradable.

The effective ask P_i^A is formally defined as:

$$P_i^A = \begin{cases} a_{1,i} & \text{if } v(a_{1,i}) > 5 \\ a_{2,i} & \text{if } v(a_{1,i}) \leq 5 \text{ and } [v(a_{1,i}) + v(a_{2,i})] > 5 \\ \emptyset & \text{otherwise (illiquid)} \end{cases} \quad (1)$$

A symmetric function defines the effective bid P_i^B using $b_{1,i}$ and $b_{2,i}$. We drop any contract i where $P_i^A = \emptyset$ or $P_i^B = \emptyset$, reducing our universe to N' tradable contracts.

For these tradable contracts, we define the market mid-price vector $\mathbf{M} \in \mathbb{R}^{N'}$:

$$M_i = \frac{P_i^A + P_i^B}{2} \quad (2)$$

We then construct the current State Matrix $\mathbf{X} \in \mathbb{R}^{N' \times F}$, where F is the number of features. Each row $\mathbf{x}_i$ represents the state of contract i :

$$\mathbf{X} = \begin{bmatrix} S_t & K_1 & \tau_1 & P_1^B & P_1^A & \sigma_{IV,1} \\ S_t & K_2 & \tau_2 & P_2^B & P_2^A & \sigma_{IV,2} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ S_t & K_{N'} & \tau_{N'} & P_{N'}^B & P_{N'}^A & \sigma_{IV,N'} \end{bmatrix} \quad (3)$$

1.2 The Fair Value Transformation

We apply a transformation function $f(\mathbf{X})$ to map the state matrix to a theoretical "fair value" vector $\mathbf{P}^{fair} \in \mathbb{R}^{N'}$. This function f represents the trained model (e.g., a learned matrix transformation, a regression model, or a gradient-boosted tree framework).

$$\mathbf{P}^{fair} = f(\mathbf{X}) \quad (4)$$

1.3 The Mispricing Score Vector ($\mathbf{W}$)

The raw mispricing is the deviation of the fair value from the market mid-price: $\mathbf{Z} = \mathbf{P}^{fair} - \mathbf{M}$. To ensure the mispricing is economically viable to trade, we normalize it by the effective bid-ask spread.

Let $\mathbf{S} \in \mathbb{R}^{N'}$ be the spread vector where $S_i = P_i^A - P_i^B$. We define the final **mispricing score vector** $\mathbf{W}$ via element-wise (Hadamard) division:

$$\mathbf{W} = (\mathbf{P}^{fair} - \mathbf{M}) \oslash \mathbf{S} \quad (5)$$

- If $W_i > 0$: The option is **underpriced** (Long candidate).
- If $W_i < 0$: The option is **overpriced** (Short candidate).
- The magnitude $|W_i|$ represents the strength of the signal relative to execution friction.

2 Part 2: Portfolio Optimization and Execution

With the score vector $\mathbf{W}$ established, we must determine the optimal position sizing for each contract. Let $\Theta \in \mathbb{R}^{N'}$ represent the target position vector, where Θ_i is the number of contracts to hold (positive for long, negative for short).

2.1 The Objective Function

The goal is to maximize the portfolio's total mispricing capture. Mathematically, this is the dot product of the position vector and the mispricing score vector $\mathbf{W}$.

$$\max \left(\Theta^\top \mathbf{W} \right) \quad (6)$$

2.2 Constraints

To ensure the resulting portfolio is safe and actionable, the maximization is subject to several linear constraints.

1. Lower Directional Exposure (Delta Constraint):

We do not require perfect delta neutrality, but rather bound it within an acceptable risk tolerance ϵ . Let $\Delta \in \mathbb{R}^{N'}$ be the vector of individual option deltas.

$$-\epsilon \leq \Delta \leq \epsilon \quad (7)$$

2. Capital Constraint:

Let $\mathbf{C} \in \mathbb{R}^{N'}$ be the vector of capital requirements (margin usage or premium paid) per contract. The total capital deployed cannot exceed the maximum allocated capital C_{max} . Since short positions consume margin and long positions consume cash, we evaluate the absolute capital requirement:

$$\|\mathbf{C}\| \leq C_{max} \quad (8)$$

3. Liquidity Bounds:

A theoretical model might suggest buying 1,000 contracts of a highly mispriced option, but the order book cannot support it. We constrain the absolute position size by the available volume $\mathbf{V}^{avail}$ established in our effective pricing logic:

$$-\mathbf{V}^{avail} \leq \Delta \leq \mathbf{V}^{avail} \quad (9)$$

Specifically, $\Delta_i \leq v(a_{1,i}) + v(a_{2,i})$ for longs, and $\Delta_i \geq -(v(b_{1,i}) + v(b_{2,i}))$ for shorts.

2.3 Summary of the Optimization Problem

The complete system solved at every 500ms tick translates to the following Linear Programming (LP) formulation:

$$\begin{aligned} \text{Maximize: } & \Delta^T \mathbf{W} \\ \text{Subject to: } & -\epsilon \leq \Delta \leq \epsilon \\ & \|\mathbf{C}\| \leq C_{max} \\ & -\mathbf{V}^B \leq \Delta \leq \mathbf{V}^A \end{aligned}$$

Solving this determines the exact quantity of each overpriced and underpriced option to trade, creating a synthesized, execution-ready arbitrage portfolio.